

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built intelligent search on *Elasticsearch* returning results across **600K+** debate documents in **under 1 second**
- Enhanced how users access information with recency-based ranking so the current month's debate topic ranks first
- Participated in *Agile* team practices with **10 engineers** on a SaaS platform drawing **10K+ site visitors**
- Improved search performance on *Elasticsearch*, returning results in **under 1 second** across **600K+** documents

Product Development Intern | Jetson

January 2025 – February 2025

- Improved user experiences by testing **3 experimental features** through **7 iterations of UI/UX testing**
- Documented findings in a **5-page report** that recommends ways to improve user engagement, delivered to the CEO
- Redesigned onboarding in *Figma* to cut it from **12 to 5 pages**, then validated the improved flow with **A/B testing**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed **200+ TikTok posts** spanning **3 years** with *Hootsuite*, *Microsoft Excel*, *Power BI* and *Python (pandas)*
- Found through data analysis that “science fun facts” posts earned **69.3%** of views from just **7%** of all content
- Recommended a “fun facts” strategy the Executive Director adopted, improving user engagement **24%** (8K+ views)

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built a **Retrieval-Augmented Generation (RAG)** agent on *Claude Sonnet 4.5* with session-scoped vector docs
- Grounded the agent's answers in session-scoped vector documents retrieved through *Supernemory* for each query
- Modeled paper data in a *PostgreSQL* database on *Supabase*, queried with *SQL* by field, year and citation count
- Integrated external *OpenAlex* and *Semantic Scholar* APIs into a data pipeline that backfills paper metadata

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Applied prompt engineering with a *JSON schema* that guides a *Claude Sonnet 4.6* agent to read truck photos
- Improved AI responses by refining the schema after wrong predictions until **10 consecutive images** were right
- Audited data quality across **15,000** truck-listing images by de-duplicating listings and filtering non-USD prices
- Built an analytical *quantile regression* model that estimates truck prices with **87% accuracy** on **30 samples**
- Won **2nd Place at UIUC Founder's Hackathon** leading team of **4** on pricing algorithm for Kamion (YC S22)

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a natural language interface where a *Gemma* Voice LLM delivers steps from **4 Random Forest** models
- Wrote a *Python FastAPI* back end that ingests water, waste and energy data from all **3 tools** to detect anomalies
- Developed **3 edge AI tools** that collect data on a school's energy, water and food use for anomaly detection
- Directed a team of 5 from ideation and development to the final presentation. **USAII Global Hackathon Finalist**

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Trained an *EfficientNetV2-S* model (**4.39M** parameters) in *PyTorch* on **15K+ images** for **96.13% accuracy**
- Published the work as an *arXiv* paper with **2 datasets** on *Kaggle*. **2nd at Arizona State Science Fair**
- Engineered a smart bin prototype on an *Arduino Uno* that sorts each waste item with **under 2 seconds** of latency
- Built a synthetic data pipeline (*U2-Net* segmentation, alpha-compositing) producing **9K** contamination images

SKILLS

Programming Languages: Python, SQL, Java, JavaScript, C++, HTML/CSS

Frameworks/Platforms: AWS, PostgreSQL, Supabase, FastAPI, Next.js, Vercel, Elasticsearch, React, Node.js, Flask

AI/ML: Claude API, Hugging Face, Vercel AI SDK, Supernemory, Gemma 4, PyTorch, scikit-learn, TensorFlow

DevTools: Git, GitHub, Claude Code, Codex, CI/CD, Docker, VS Code, Figma, Bun, Clang, Catch2, Arduino IDE

Concepts/Practices: Prompt Engineering, Large Language Models, Generative AI, RAG, Vector Search, Embeddings